ELSEVIER

Earth and Planetary Science Letters 203 (2002) 353–367

EPSL

www.elsevier.com/locate/epsl

New paleomagnetic result from the Ethiopian flood basalts in the Abbay (Blue Nile) and Kessem gorges

Tesfaye Kidane [a,b,*], Bekele Abebe [b], Vincent Courtillot [c], Emilio Herrero [a]

[a] *SOEST, Hawaii Institute of Geophysics and Planetology, University of Hawaii at Manoa, Paleomagnetic Laboratory, 2525 Correa Road, Honolulu, HI 96822, USA*

[b] *Department of Geology and Geophysics, Addis Ababa University, P.O. Box 1176, Addis Ababa, Ethiopia*

[c] *Laboratoire de Paléomagnetisme et de Tectonique, Institut de Physique du Globe de Paris, 75230 Paris Cedex 05, France*

Received 7 January 2002; accepted 11 July 2002

Abstract

New paleomagnetic investigations on the Ethiopian trap series have been undertaken at the Abbay and Kessem gorges in an attempt to better constrain the 30 Ma paleomagnetic pole of Africa. We sampled six thick massive basaltic lava flows, totaling 230 m, from Abbay Gorge and 10 lava flows, 180 m in thickness, from Kessem Gorge. Detailed paleomagnetic analyses disclosed that the carriers of the characteristic remanent magnetization (ChRM) are different in different lava flows. These are mostly titanomagnetites, titanomaghemites, and magnetite minerals with a broad range of coercive force and blocking temperatures. The heating and cooling susceptibility vs. temperature curves, many of which are irreversible, may indicate chemical remagnetization, notably low temperature maghemitization. Only one flow (KS04) with a clear 580°C Curie temperature was apparently unaffected by chemical remagnetization. The ChRM direction of this flow is identical to that in other flows, which suggests that if and when remagnetization occurred, this was shortly after emplacement of the lava flows. All of the flows sampled have normal polarity. However, a reversed component of low to medium coercive force and low to medium unblocking temperature occurs in flow KS01 at Kessem Gorge. The ChRM directions for the 16 sites are $D = 3.1°$, $I = 5.8°$ ($\alpha_{95} = 12.7°$). The paleomagnetic pole obtained from these is at $\lambda = 83.0°$N, $\phi = 193.3°$E ($A_{95} = 9.0°$). Comparison with three previous studies of the traps shows remarkable consistency and a number of means are derived and discussed. Two final preferred poles for the traps are at $\lambda = 79.0°$N, $\phi = 196.9°$E ($A_{95} = 2.8°$) when all 112 published flows are used, and $\lambda = 78.7°$N, $\phi = 209.4°$E ($A_{95} = 3.4°$) when only the 76 flows from the four more recently analyzed sections are included. Both are compatible with the recent reference synthetic pole for Africa of Courtillot and Besse [J. Geophys. Res. (2002) in press]. In that sense, the Ethiopian trap pole is not anomalous and does not require more of a non-dipolar contribution than indicated by analyses of the global paleomagnetic data base covering the last few million years.
© 2002 Elsevier Science B.V. All rights reserved.

Keywords: Ethiopia; Oligocene; pole positions; flood basalts; paleomagnetism

1. Introduction

Though the global data base of paleomagnetic results continuously increases, there are still a

* Corresponding author. Fax: +251-1-550655.
E-mail addresses: dgg@telecom.net.et (T. Kidane), tesfayek@yahoo.com (T. Kidane).

0012-821X/02/$ – see front matter © 2002 Elsevier Science B.V. All rights reserved.
PII: S0012-821X(02)00834-8

number of continents and periods for which data are scarce and using a predicted, or synthetic, pole based on data from other continents can be useful [1,2]. However, such synthetic curves require constant checking and acquisition of new data. With this goal in mind, we propose new data from stable Africa at 30 Ma ago. The Ethiopian trap series, known to represent the youngest large igneous province in the world (Fig. 1), covers $\sim$600 000 km^2 in surface area excluding once contiguous Yemenite volcanics [3]. Its thickness reaches up to 2 km with an estimated total volume of $\sim$350 000 km^3 excluding the unknown volume of the traps beneath the Afar and the East African rift valley (and the significant part which was likely removed by erosion [4]).

Recent ^{40}Ar/^{39}Ar studies carried out on this series at different places in the NW Ethiopian Plateau reported the age to be 30 Ma with a duration of about 1 Ma [5,6]. The earliest paleomagnetic studies performed on the traps date back to 1970 [8]. They had obtained good results both at the Blue Nile Gorge (BNG) and at Mugher Canyon (MC), whereas at Kessem valley section the formations they sampled were later found to be Plio–Pleistocene in age [9]. The Blue Nile Gorge section of Brock et al. [8] was sampled from the Goha Tsion side of the Blue Nile Gorge (i.e. opposite of the present study).

Detailed magnetostratigraphies at Lima Limo and Wegel Tena, by Hofmann [5,6] and Rochette et al. [7], are, so far, the best-constrained paleomagnetic studies. The poles calculated from these paleomagnetic studies have been reported to be far-sided by 6° ± 6°, when compared to the reference synthetic pole for Africa [1,2,10] and yielded large paleosecular variation [7]. Residual remagnetizations could be a reason for the observed discrepancy. In order to improve the reliability of the pole position for this voluminous Oligocene trap, as many spatially distributed sites as possible need to be sampled, in order to take into consideration lateral variation of the formation and hopefully ensure that potential remagnetizations have been identified and removed.

The main objective of this work was therefore to perform new paleomagnetic studies on the trap series at remote geographic locations. To address this objective, paleomagnetic samplings were carried out at two selected locations where previous geochronologic age determinations were already available.

2. Sampling

The two locations where sampling was undertaken are the Abbay and Kessem gorges (Fig. 1).

The Abbay (or Blue Nile) Gorge section (ABG) is located at 10°9′N latitude, 38°10′E longitude. Six thick lava flows, totaling 230 m in thickness, are exposed along a road cut on the main Addis Ababa–Debre Markos road. These massive lavas, chiefly basaltic, unconformably overlie the Oxfordian–Kimmeridgian marl–limestone intercalation [11]. These flows are generally cliff-forming units, each separated by gentle slope-forming layers of volcanic ash, 8–20 m thick. The ^{40}Ar/^{39}Ar age determinations performed at the base of this conformable lava flow section indicate ages of 29.6 ± 0.1 Ma [6].

The Kessem Gorge section is located at 9°10′N latitude, 39°30′E longitude on the southeastern margin of the western plateau at the bottom of Kessem River, on both sides of a new bridge. Eleven different units, ranging in composition from basalt to rhyolites, with a total thickness of 180 m, are exposed within the river; all are tilted at 060°/30°SE. An altered and friable volcanic breccia (not sampled) about 50 m thick separates the lowermost basaltic unit from the second unit. The remaining nine top flows conformably overlie the second unit and the whole succession is unconformably overlain by $\sim$300 m thick flat-lying volcanic formations (loosely welded tuffs and ignimbrites) of Pliocene age [9]. Most of the cores sampled were oriented in the field with both solar and magnetic compasses. The difference between the two readings is insignificant.

There is no recent direct age determination to date on the flows we sampled in the Kessem Gorge. However, conventional K/Ar age determinations by Rex et al. [9] had obtained ages of 20.9 ± 0.5 Ma for the lower flat-lying ignimbrite unit, unconformably overlying the 11 flows sampled and giving an upper age limit. When

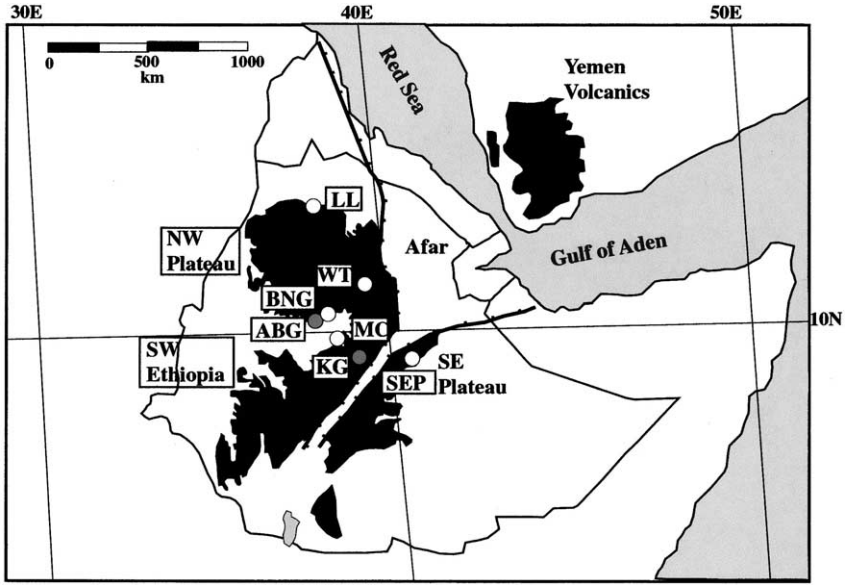


Fig. 1. Generalized location map of the studied Ethiopian plateau in relation to the Afar depression and East African rift. The full circles show the positions of present (gray) and previous studies (white). LL = Lima Limo, WT = Wegel Tena, MC = Mugher Canyon, SEP = South East Plateau, ABG = Abbay Gorge, BNG = Blue Nile Gorge, KG = Kessem Gorge.

new decay constants by Steiger and Jäger [12] are used, this age can be recalculated to give 21.5 Ma. Considering the unconformity between these two sequences, the 10 flows we sampled must be the equivalents of the Ethiopian trap series dated elsewhere at 30 Ma.

3. Laboratory treatment and data analyses

The collected core samples were cut in the laboratory into 2.2 cm long standard specimens. Part of the measurements was done in the shielded room of the paleomagnetic laboratory at the Institut de Physique du Globe de Paris. The other part was done in the recently rebuilt double-wall mu-metal and transformer steel magnetostatic shielded room of SOEST-HIGP paleomagnetic laboratory in Hawaii, USA. In both laboratories magnetic measurements were performed with 2G Cryogenic magnetometers (horizontal access, with online alternating field (AF) demagnetization with maximum peak field of 100 mT) and JR5 Spinner magnetometers. AF demagnetization was also done with single-coil Schönstedt systems. The rock magnetic experiments were done using a KLY2 (Kappa bridge) instrument with a low-field susceptibility vs. attachment to temperature determine Curie points (−200 to 700°C), a 1 T electromagnet, and a laboratory-built impulse magnetizer for isothermal remanent magnetization (IRM) studies.

4. Paleomagnetism

In both Abbay and Kessem sections, eight to 10 cores were generally collected from each flow; however, where flows were very difficult to drill, a larger number of shorter cores were collected. Thermal (Th) and AF stepwise (11–28 steps) demagnetizations were performed on specimens from each core, in parallel fashion for all sites, in order to detect any remagnetization effect. Demagnetization results from the treated samples indicate that magnetic behavior at the two sections is different.

The Th cleaning process mostly eliminates a viscous remanent magnetization (VRM) between natural remanent magnetization (NRM) and 150–

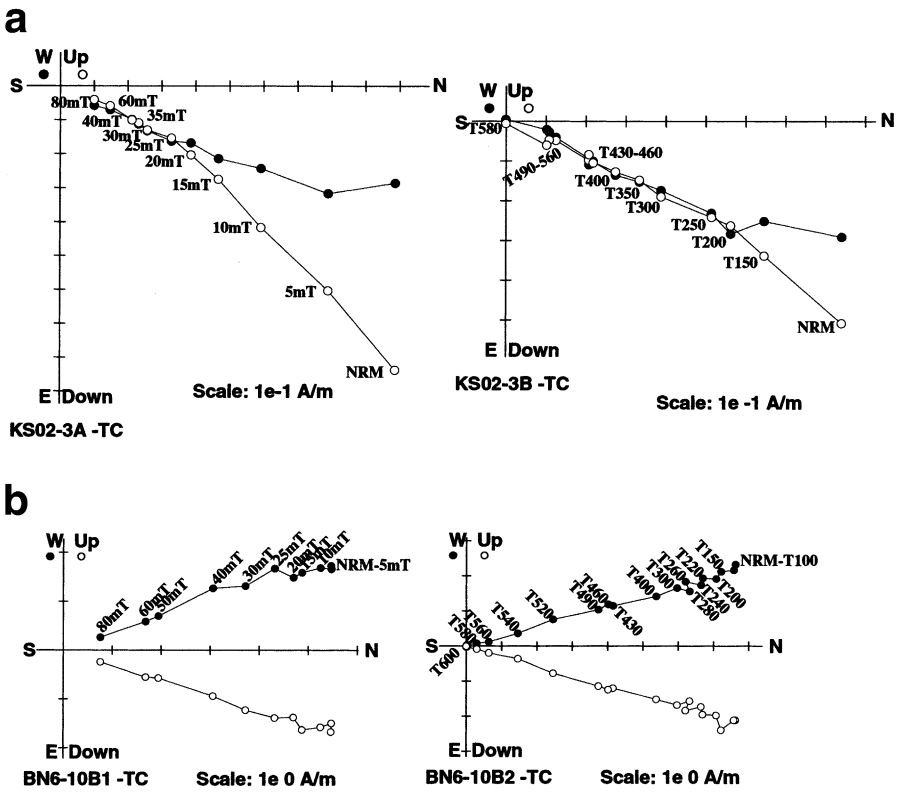


Fig. 2. Zijderveld diagrams for typical specimens: (left) AF demagnetization (in mT) and (right) thermal demagnetization (temperatures in °C). Specimens shown on the same line are from the same core. (a) From the Kessem Gorge. (b) From the Blue Nile Gorge.

200°C steps (e.g. Fig. 2), but for cores from site KS01 the VRM required a heating to 300–350°C (e.g. Fig. 3). Above these temperatures the characteristic remanent magnetization (ChRM) is identified and isolated (Figs. 2 and 3). Three different demagnetization characteristics were observed. These are a steady intensity decay from NRM through the final steps varying between 520 and 580°C (e.g. KS02-3B, Figs. 2a and 4), a steady decrease from NRM up to 490°C with a sharp drop at 500–560°C (e.g. BN6-10B2, Figs. 2b and 4) and complete demagnetization by 300°C (e.g. KS10-7A, Fig. 4). For most studied samples, one or two components of magnetization are found, the first being a VRM acquired in the recent geomagnetic field. For KS01, one to three non-overlapping components of magnetization were found upon complete demagnetization. The first component (low unblocking temperature) is isolated between NRM and 250°C and is normal in polarity. The second component (medium unblocking temperature), isolated between 250°C and 350°C, is reversed in polarity. The third component (high unblocking temperature), isolated between 350°C and 580°C, has normal polarity and is directed to the origin. The high and low unblocking temperature components have paleomagnetic directions parallel to one another, which in turn are anti-parallel to the reversed medium unblocking temperature component.

The AF experiment results, like the Th, indicate a simple demagnetization behavior with one or two components of magnetization for most of the samples. The first one, often cleaned between NRM and 5–20 mT, is a VRM. Demagnetization steps above 25 mT generated a straight line directed to the origin and were interpreted as the ChRM. For KS01, three components of magnet-

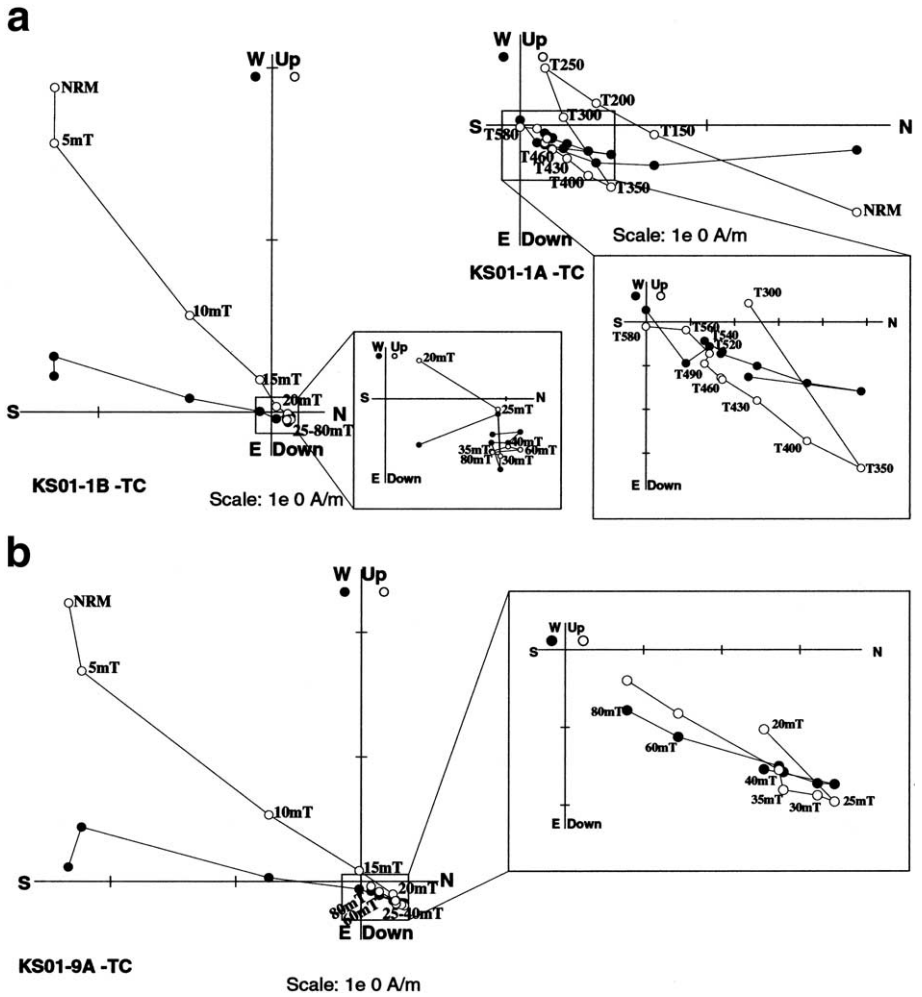


Fig. 3. Zijderveld diagrams for specimens from flow KS01: (left) AF demagnetization (in mT) and (right) thermal demagnetization (temperatures in °C). Specimens shown on the same line are from the same core. (a) Specimen where the third component was not isolated by AF. (b) Specimen where the third component was isolated.

ization were identified. A soft present field remagnetization is removed by 5 mT. The second one is isolated between 5 and 25 mT and has reversed polarity while the third component is isolated between 40 and 80 mT after a strong resistance to demagnetization between 25 and 40 mT (Fig. 3b). For some of the specimens the resistance is so high that this component was not isolated at all at 80 mT, though it is consistent with the magnetization direction above 80 mT (Fig. 3a). The third component, when isolated, is directed to the origin, and interpreted as the ChRM.

5. Th vs. AF demagnetization results

For most flows, both AF and Th diagrams show identical results (Fig. 2), with two components being resolved. The first component is isolated between NRM and 25 mT or 200°C. The magnetization above these steps is often aligned toward the origin and defines the ChRM (Figs. 2 and 3). However, the demagnetization behaviors encountered at some sites in Kessem Gorge are worth mentioning. For instance, demagnetization levels above 25 mT are unable to isolate the last

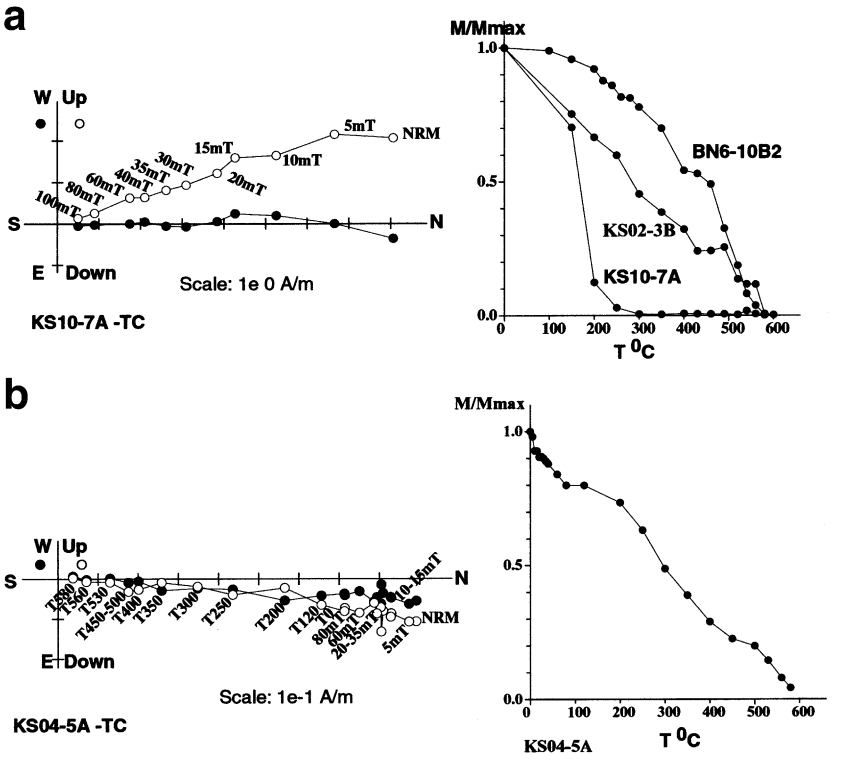


Fig. 4. (a) Representative magnetic moment decay curve pattern of three selected samples from both sections and a Zijderveld diagram for sample KS10-7A. (b) Zijderveld diagram and associated magnetic moment decay curve of sample KS04-5A, presenting strong resistance to AF but completely demagnetized at 580°C.

10–20% of the total magnetization for specimen KS01-1B (Fig. 3a) and this component was isolated only after the 40 mT step for sample KS01-9A (Fig. 3b) indicating that grains with a high coercivity spectrum (HCS) carry the normal polarity component. An experiment performed using a Th technique on the same twin samples likewise revealed three components: a normal component with low unblocking temperatures (LT) between NRM and 250°C, a normal component with medium unblocking temperatures (MBT) between 250 and 350°C, and a normal component with high unblocking temperature (HT) between 350 and 580°C (Fig. 3). The LT and HT components are parallel. This indicates that the magnetic minerals with HCS are also characterized by LT and HT, and that the component with both MBT and a medium coercivity spectrum might be the product of a (thermo-)chemical remanent magnetization (CRM) linked to emplacement of the overlying friable volcanic breccia, which was not sampled.

Another sample worth mentioning is KS04. This sample showed a very strong resistance to AF demagnetization, retaining more than 70% of its magnetization at 80 mT. A Th technique was then used to clean the rest of magnetization starting from 100°C through 580°C. In addition, on its twin specimen, the ChRM was successfully isolated using only the Th technique (Fig. 4b). This illustrates that different magnetic minerals (likely fine-grained titanomagnetite and possible some titanohematite with a wide range of unblocking temperatures going from 100 to 600°C) carry the same magnetization in this particular flow.

The median destructive field varies from 5 to 25 mT, except for sample KS04 for which only 30% of the total magnetization was isolated at 80 mT (Fig. 5). The various types of decay of

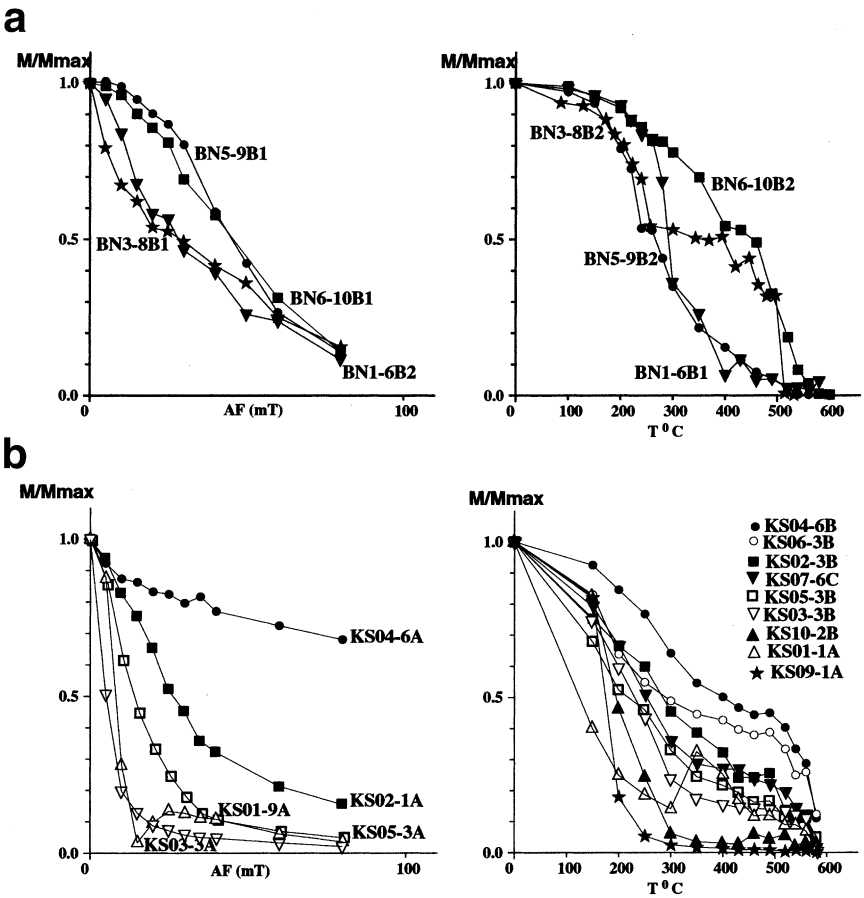


Fig. 5. Normalized intensity curves for AF (left) and Th demagnetization. (a) Behavior in Blue Nile Section samples. (b) Behavior in the Kessem Gorge samples.

magnetic moments as a function of temperature and AF demagnetization steps are illustrated in Fig. 5.

6. Susceptibility vs. temperature (*K*–*T*) experiments (Fig. 6)

Experiments performed on representative samples in both gorges reveal several ranges of transition temperatures, which could be attributed to distinct magnetic minerals or, when they are not reversible, to chemical transformation during heating in the laboratory. These changes occur in various temperature ranges, depending on samples and sections, between 150 and 300°C, 450 and 500°C and 540 and 600°C (Abbay, Fig. 6a,b) and between 300 and 320°C, 400 and 450°C, and 530 and 580°C (Kessem, Fig. 6c,d). Increases in K_{tot} are sometimes observed (above 320°C, Fig. 6a, or 420°C, Fig. 6b). The final K_{tot} of most samples is higher than the initial K_{tot} before heating, possibly due to maghemitization [13]. The acquisition of IRM shows saturation below 1 T, indicating a soft mineral, possibly titanomagnetite (Fig. 7). These curves could indicate the presence of titanomaghemite, titanomagnetite, and magnetite. In samples such as KS04 (Fig. 6d), reversible curves and a single Curie temperature at 580°C indicate that no change in mineralogy occurred and the direction is likely the ChRM acquired during original cooling. In other cases, magnetization could be a CRM due to low temperature maghemitization. More than half of

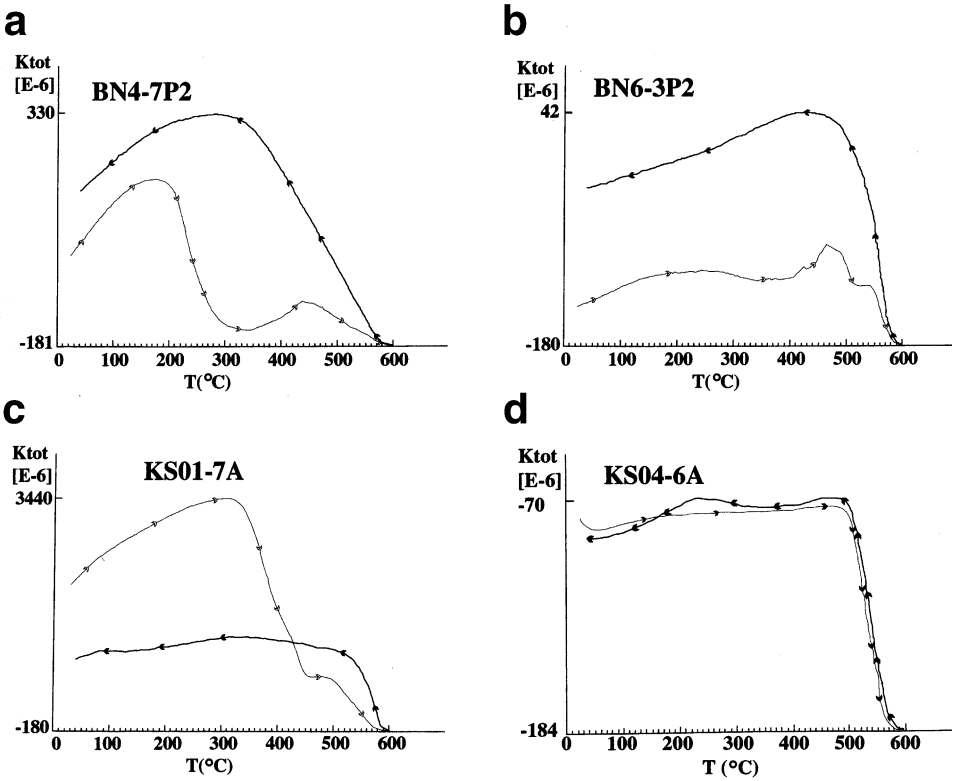


Fig. 6. Representative susceptibility vs. temperature (K–T) diagrams from the Abbay Gorge (a, b), and Kessem Gorge (c, d). The heating and the cooling curves are shown by lines in gray and black respectively, and also shown by the arrowheads. The two curves are irreversible in a–c and reversible in d.

the samples used for paleointensity study by [14] failed to yield reliable estimates, which might be due to such remagnetization. The crossover point of the IRM acquisition and demagnetization curves ranges from 0.5 to 0.65 (Fig. 7). These crossover values indicate grain sizes from single to pseudo single domain [15]. The directions of the components (CRM or ChRM) carried by minerals affected in the critical temperature ranges seen in Fig. 6 are similar to the ChRM directions when these can be documented (such as in Fig. 6d). We therefore assume that the directions we have isolated in most sites are indeed characteristic directions representative of early NRM acquisition.

7. Mean paleomagnetic directions

After having isolated the ChRM, principal component analysis was performed following Kirschvink [16]. A maximum angular deviation less than 1° characterizes most ChRM directions. All paleomagnetic components were calculated at the specimen level, results of specimens belonging to the same core were first averaged and a mean for the site or flow was then calculated, taking each core as a unit. Table 1 reports the mean paleomagnetic directions obtained at the two sections.

In Abbay Gorge, all six flows are found to have normal polarity with one of the sites having a negative inclination. The confidence limit α_{95} for each site is less than 9° for five of the six sites and is 10.1° for one of the sites, BN05 (Table 1). Due to the small number of flows sampled, site means are scattered and the Abbay section mean has large 95% uncertainty. The mean paleomagnetic directions for both stratigraphic and tilt-corrected (Fig. 8a) coordinates are of course the same

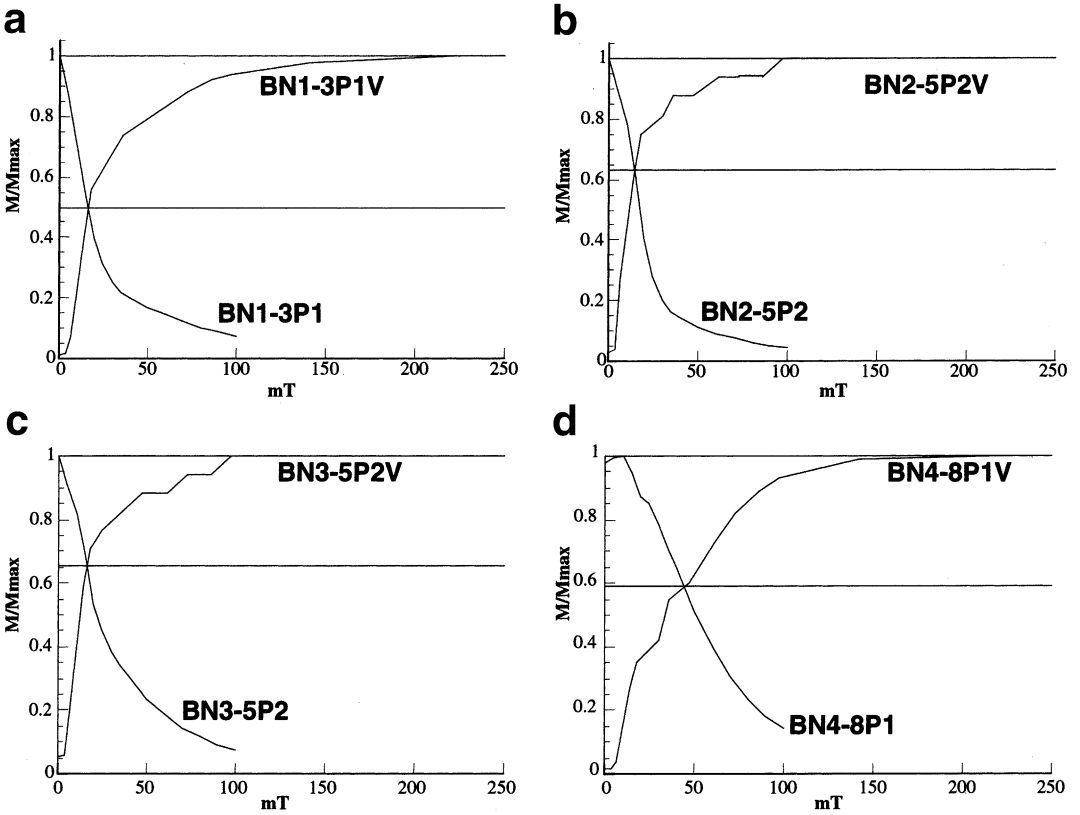


Fig. 7. Four representative IRM acquisition curves and corresponding AF demagnetization of same specimens. Th acquisition curves rise steeply and reach saturation well before 0.5 T, indicating that there is no contribution to the IRM from hematite.

since flows are flat-lying: $D = 351.7°$, $I = 16.9°$ ($\alpha_{95} = 22.0°$, $n = 6$).

At Kessem Gorge the first flow KS01 has normal polarity, with a significant reversed remagnetization. The 50 m thick friable volcanic breccia, not sampled for paleomagnetism, which overlies this flow may have been emplaced at a time of reversed polarity and may have remagnetized it. Both the normal and reversed directions are well isolated in most samples and are stable. They are also found to be anti-parallel. Considering that these represent snapshots of the geomagnetic field direction at two different but close time periods, we report both of them as separate results. Whether we take only the normal polarity or both does not significantly change the overall mean. As a result both directions from this single flow are listed in Table 1 but only the normal component is included in the overall mean. The nine flows above have normal polarity. Eight of the 11 means have α_{95} less than 11° and the remaining three have α_{95} between 11 and 14°. Formations sampled at Kessem Gorge are uniformly tilted at 060°/30°SE. As a result, the mean directions in stratigraphic and tectonic (Fig. 8a) coordinates are respectively: $D_g = 14.0°$, $I_g = -23.5°$ ($\alpha_{95} = 14.8°$; $n = 10$) and $D_s = 9.6°$, $I_s = -0.9°$ ($\alpha_{95} = 14.8°$; $n = 10$) (see also Table 1 and Fig. 8).

The site mean directions from the Kessem and Abbay gorges are given in Table 1 before and after tectonic correction (and Fig. 8b). Despite rather large 95% uncertainties, it is clear that their confidence intervals do not intersect prior to correction, but do afterwards. The angular difference between the Abbay and Kessem gorge means after tilt correction is $16.9 \pm 27.5°$. The two populations share the same precision parameter at the 95% confidence level, and the two means are dis-

Table 1
Site (flow) paleomagnetic mean directions averaged after AF and Th demagnetization for all of the sites in the Abbay and Kessem sections

Sample id.	Strike	Dip	n	D_g	I_g	D_s	I_s	K	α_{95}
Abbay (Blue Nile) Gorge									
BN01	0	0	8	340.7	−18.7	340.7	−18.7	66.8	6.8
BN02	0	0	7	355.7	40.2	355.7	40.2	49.0	8.8
BN03	0	0	15	2.9	27.1	2.9	27.1	22.9	8.2
BN04	0	0	11	18.2	17.5	18.2	17.5	48.5	6.6
BN05	0	0	13	338.6	8.1	338.6	8.1	17.7	10.1
BN06	0	0	12	336.2	22.5	336.2	22.5	39.4	7.0
BN (ABG) mean	0	0	6	351.7	16.9	351.7	16.9	10.2	22.0
Kessem Gorge									
KS01R	060	30	7	176.4	−13.2	184	−39.3	22.9	13.0
KS01N	060	30	5	17.8	5.7	24.3	24.9	59.8	10.0
KS02	060	30	8	19.7	−3.4	22.3	15.7	74.4	6.5
KS03	060	30	8	358.3	−13.2	358.4	13.4	43.8	8.5
KS04	060	30	7	4.0	−23.7	.8	1.8	32.9	10.7
KS05	060	30	9	4.6	−8.8	5.7	15.9	21.9	11.2
KS06	060	30	10	33.6	−50.8	11.9	−32.1	26.9	9.5
KS07	060	30	10	9.0	−20.2	6.3	3.8	22.0	10.5
KS08	060	30	4	29.4	−57.9	4.8	−36.7	142.7	8.2
KS09	060	30	9	26.3	−13.0	24.4	4.3	15.0	13.7
KS10	060	30	9	8.8	−48.2	356.9	−22.7	70.8	6.2
KG mean	060	30	10	14.0	−23.5	9.6	−0.9	11.6	14.8
Overall mean			16	3.5	−8.7			6.4	15.9
						3.1	5.8	9.4	12.7

The latitude and longitude of the Abbay and Kessem gorges are 10°9′N, 38°10′E and 9°10′N, 39°30′E respectively. Sites KS01R and KS01N correspond to the mean directions of the reversed and normal components of flow KS01. Sample id., site code number; Strike and Dip of flow; n, number of samples; D_g and I_g, declination and inclination in geographic coordinates; D_s and I_s, declination and inclination in stratigraphic coordinates; K, Fischer precision parameter; $\alpha 95$, 95% confidence interval.

tinct before and identical after correction, at the same level [17]. Therefore, a common overall mean can be calculated by combining all site mean values together. One finds (Fig. 8c and Table 1): $D_g = 5.5°$, $I_g = -8.7°$ ($\alpha_{95} = 15.9°$; $n = 16$) and $D_s = 3.1°$, $I_s = 5.8°$ ($\alpha_{95} = 12.7°$; $n = 16$). The above discussion leads us to propose that the latter is the primary magnetization.

8. Paleosecular variation and virtual geomagnetic poles (VGP)

The Kessem and Abbay gorge mean paleomagnetic pole positions (Table 2) can be combined to yield an overall mean paleomagnetic pole position for our study. This pole is located at $\phi = 193.3°$E, $\lambda = 83.0°$N ($n = 16$, $A_{95} = 9.0°$) (Fig. 9a).

We can next compare this pole to previous ones obtained for the Ethiopian traps. Two studies date from the early 1970s [8,18]. Brock et al. [8] sampled three sections, only two of which appear to be coeval with the traps: the Blue Nile Gorge and Mughar Canyon sections, both located within the main Ethiopian plateau. Brock et al. sampled only one or two blocks per flow, one flow being considered as a site. The specimens were demagnetized in AF up to 60 mT and measured on an astatic magnetometer, and the direction after 5–10 mT was generally retained. Ten flows (27 blocks) yielded characteristic magnetizations for Mughar Canyon and four flows (13 blocks) for Blue Nile Gorge. The corresponding section mean poles are given in Table 2. Schult [18] collected 119 cores from 30 sites on the northwestern margin of the Somalian part of the traps, the so-called southern plateau. Remanent magnetization was measured with a fluxgate spinner and samples were subjected to AF demagnetization up to 10–40 mT. Schult finally retained 11 normal and

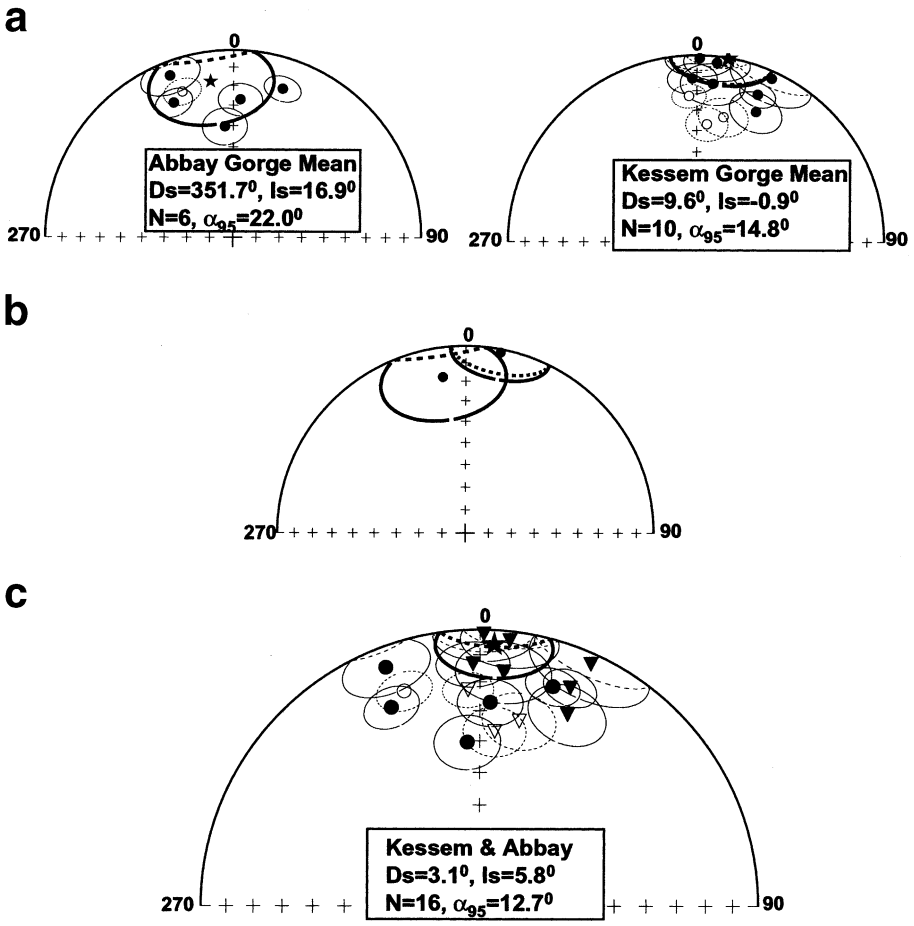


Fig. 8. Stereographic projections. (a) Site means from the Abbay (left) and Kessem (right) gorges, together with their respective means. (b) the mean directions from Kessem and Abbay gorges compared. (c) The overall mean directions for the sites from both sections.

11 reversed sites and his overall mean pole is given in Table 2. Much more recently, a thorough paleomagnetic and geochronologic study of two sections has been published [5–7]. Methods are very similar to those used in our study. Rochette et al. [7] drilled six to 10 cores in each site, taken to be a single cooling unit. The two sections from the main part of the Ethiopian plateau are located at Wegel Tena and Lima Limo. Both AF and thermal demagnetization were performed, and Zijderveld diagrams and principal component analyses were used to analyze the data. The two sections yielded statistically similar poles that were averaged together. The separate mean poles are given in Table 2. The five section poles from the literature and the two new ones obtained in our paper are shown in Fig. 9. All have 95% confidence intervals less than 15° (a cutoff level often used in selection of acceptable means), except for ABG which has a confidence interval of 22.0°) linked to the smaller number of flows (6). Six mean poles are rather well grouped (and the seventh is compatible given its confidence interval) and one can calculate an overall average, with or without section ABG. The results are respectively: $\phi = 193.1°$E, $\lambda = 81.6°$N ($n = 7$ sections, $A_{95} = 6.0°$) and $\phi = 185.9°$E, $\lambda = 79.7°$N ($n = 6$ sections, $A_{95} = 4.9°$), and are virtually identical.

One might worry about averaging sections with vastly different numbers of sites (four to 40) or

Table 2
Mean paleomagnetic poles for the Ethiopian traps based on different sections from the literature and this study, on combinations of section means, on combinations of individual flow mean directions, and a synthetic 30 Ma reference pole for Africa [2]

Section	n	ϕ_s	λ_s	A_{95}	Reference
Section means					
BNG	4	147.8	84.4	11.9	[8]
MC	10	180.3	80.8	5.8	[8]
SEP	22	170.2	75.1	6.4	[18]
LL	40	207.2	75.8	4.7	[7]
WT	20	225.8	81.0	5.6	[7]
ABG	6	301.2	81.9	16.7	This study
KG	10	175.2	76.2	9.5	This study
Combined section means					
Mean (LL to KG)	4	213.4	81.4	10.4	This study
Mean (without ABG)	6	185.9	79.7	4.9	This study
Mean (all)	7	193.1	81.6	6.0	This study
Site/flow/cooling unit means					
Mean (ABG, KG)	16	193.3	83.0	9.0	This study
Mean (ABG, KG, LL, WT)	76	209.4	78.7	3.4	This study
Mean (all seven sections)	112	196.9	79.0	2.8	This study
Besse and Courtillot 30 Ma synthetic pole for Africa					
BC02	24	192.0	81.5	3.7	[2]

obtained with very different techniques (two in the 1970s, two almost 30 years later). For instance, our results from Abbay Gorge and previous results from other parts of the gorge [8] are rather different. This might be due to some unremoved recent remagnetization in the Blue Nile Gorge data. Indeed these early data were obtained at a time when current demagnetization procedures and tests on magnetic mineralogy and remagnetizations had not been fully developed; for instance, only AF demagnetization up to 60 mT was used. If only the four sections from the two most recent studies are averaged, the mean pole becomes $\phi = 213.4°$E, $\lambda = 81.4°$N ($n = 4$ sections, $A_{95} = 10.4°$). The pole is statistically similar to the previous two averages but has much larger uncertainty. Also, one might wish to average out individual flow or cooling units from all studies to assign a better weight to each contribution. The problem with the two early studies is that most flow means are based on only two cores, or blocks. Still, using the mean flow directions as listed in the early papers, and combining the recent studies, we obtain an overall average pole of $\phi = 196.9°$E, $\lambda = 79.0°$N ($n = 112$ flows, $A_{95} = 2.8°$). One could prefer averaging only flows from modern studies, yielding $\phi = 209.4°$E, $\lambda = 78.8°$N ($n = 76$ flows, $A_{95} = 3.4°$).

These various poles are illustrated in Fig. 9b. It is interesting that all of them are contained within the 95% confidence interval of the new synthetic pole proposed by Besse and Courtillot [2] for stable Africa at 30 Ma. The flows and sections from all studies provide two mean poles, which are just within the intersection of their 95% confidence intervals. The similar pair of poles for only the two more recent studies does the same, though in that case the 95% confidence interval for the sections is quite large. When flows are considered, the collection of 76 flows from the two modern studies and that of all 112 published flows are also within the intersection of their 95% confidence intervals. So this whole set displays a remarkable consistency. We would tend to prefer using only recent studies, which avoids the risk of undetected remagnetization components incurred in the early studies, where blanket demagnetization and lack of principal component analysis could have affected the results. We also tend to prefer averages of flows rather than sections, which weights individual field data better. So, our final preferred overall Ethiopian trap mean pole would be that

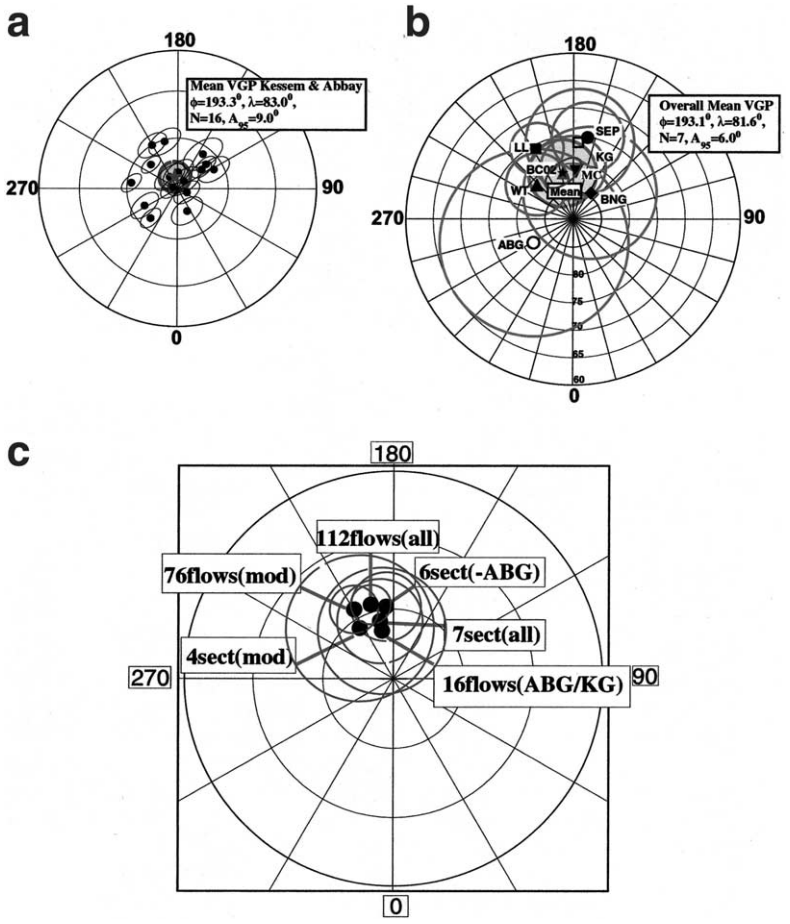


Fig. 9. Stereographic projections. (a) The Kessem and Abbay individual site VGPs and the overall VGP of the two sections of this study. (b) The mean VGPs obtained from previous work at different sections (BNG, MC, SEP, LL and WT), and the present work (ABG and KG), together with the expected synthetic pole position (BC02) for Africa at 30 Ma from [2]. The symbols (diamond, inverted triangle, full circle, full rectangle, full upright triangle, open circle, open rectangle, full white color upright triangle and star) stand for Blue Nile Gorge, Mugher Canyon, South East Plateau, Lima Limo, Wegel Tena, Abbay Gorge, Kessem Gorge, BC02, and the resulting overall mean respectively. (c) Some corresponding mean poles calculated from individual flow data (see text and Table 2).

at $\phi = 209.4°$E, $\lambda = 78.7°$N ($n = 76$ flows, $A_{95} = 3.4°$), although from the previous discussion and Fig. 9b it is clear that using other poles, such as the 112 flows mean pole, would not change the results. It is worth pointing out that the angular standard deviation for these means ranges between 16 and 17°, close to the global average extrapolated near the equator for the 22.5–45 Ma period, which is 15.4° [19]. This indicates that secular variation is correctly represented in the whole collection of lava flows and that there is no need to invoke large additional sources of uncertainty to account for data scatter.

9. Conclusion

In our paleomagnetic investigation at Abbay and Kessem gorges in the main part of the Ethiopian trap province, thermal and AF demagnetization techniques gave a relatively simple demagnetization behavior characterized by one to three

paleomagnetic directions. The *K–T* experiments suggest that the ChRM was acquired during or shortly after emplacement of the lava flow units (though possibly involving some amount of low temperature maghemitization of original titanomagnetites). Despite the rather large confidence intervals of the individual overall mean directions which is due to the small number of sites, the overall mean direction obtained by combining results from both sections gives what we believe to be a reliable result. The mean VGP calculated from this mean direction is statistically identical to the recent synthetic pole position obtained by Besse and Courtillot for Africa at the same age, by transferring data from other plates and performing a global average [2]. Our study therefore validates the synthetic curve for that time, and provides new Africa-based data which can be added to the global data base for use in future studies.

Comparison of our results with three other studies, two obtained in the early 1970s and one recently, shows remarkable overall consistency. Most section mean poles are in statistical agreement, and so are mean poles based on individual flow mean (or site mean, or cooling unit mean) results. The two overall mean poles which can be preferred (depending on whether one includes the earlier data despite less sophisticated laboratory and data analysis or not) both agree with the synthetic pole and with each other. The pole based on the more modern data ($\phi = 209.4°$E, $\lambda = 78.7°$N; $A_{95} = 3.4°$) has a statistically insignificant angular difference of $3 \pm 5°$ with respect to the synthetic pole. Recent paleointensity results from the Lima Limo section [14] report a low field strength, which is in agreement with high paleosecular variation, and in general supports our conclusions. Our results as outlined above, namely the preferred pole using flow means and recent studies, are consistent with a small quadrupolar field component as has been found for the more recent Afar basalts ($6 \pm 2\%$ of the axial dipole [20]) or for the overall global database over the last 5 million years (3–6% of the axial dipole [21]). However, this new pole does not seem to require the much larger value (14% of the axial dipole) advocated in [7].

Acknowledgements

The authors are grateful to thank Pierre Rochette and an anonymous reviewer for helpful suggestions. Field work was financed by the Department of Geology and Geophysics, AAU and INSU. *[AC]*

References

[1] J. Besse, V. Courtillot, Revised and synthetic apparent polar wander paths of the African, Eurasian, North American and Indian plates, and true polar wander since 200 Ma, J. Geophys. Res. 96 (1991) 4029–4050.
[2] J. Besse, V. Courtillot, Apparent and true polar wander and the geometry of the geomagnetic field in the last 200 million years, J. Geophys. Res. (2002) in press.
[3] P. Mohr, Ethiopian flood basalt province, Nature 303 (1983) 577–584.
[4] C. Dessert, B. Dupré, L.M. François, J. Schott, J. Gaillardet, G. Chakrapani, S. Bajpai, Erosion of Deccan traps determined by river geochemistry: impact on the global climate and the $^{87}Sr/^{86}Sr$ ratio of seawater, Earth Planet. Sci. Lett. 188 (2001) 459–474.
[5] C. Hofmann, Datation $^{40}Ar/^{39}Ar$ et paléomagnetisme des traps d'Ethiopie, du Deccan et de Siberie, PhD Thesis, Univ. Paris 7-IPGP, 1997, 200 pp.
[6] C. Hofmann, V. Courtillot, G. Feraud, P. Rochette, G. Yirgu, E. Ketefo, R. Pik, Timing of the Ethiopian flood basalt event and implications for plume birth and global change, Nature 389 (1997) 838–841.
[7] P. Rochette, E. Tamrat, G. Feraud, R. Pik, V. Coutillot, E. Ketefo, C. Coulon, C. Hoffmann, D. Vandamme, G. Yirgu, Magnetostratigraphy and timing of the Oligocene Ethiopian traps, Earth Planet. Sci. Lett. 164 (1998) 497–510.
[8] A. Brock, I.L. Gibson, P. Gacii, The paleomagnetism of the Ethiopian flood basalt succession near Addis Abeba, Geophys. R. Astron. Soc. 19 (1970) 485–497.
[9] D.C. Rex, L. Gibson, F. Dakin, Age of the Ethiopian flood basalt succession, Nature 230 (1971) 131–132.
[10] F. Torcq, Evolution et destruction de la Pangée du Carbonifère au Jurassique: dérivé des poles magnétiques et étude de la dipolarité du champ. PhD Thesis, Univ. Paris 7, 1997, 183 pp.
[11] G. Assefa, Lithostratigraphy and environment of eposition of the Late Jurassic-Early Cretaceous sequence of the central part of Northwestern Plateau, Ethiopia, N. Jb. Geol. Paläontol. Abh. (Stuttgart) 182 (1991) 255–284.
[12] R.H. Steiger, E. Jäger, Subcommission on geochronology: Convention on the use of decay constants in geo- and cosmochronology, Earth Planet. Sci. Lett. 36 (1977) 359–362.

[13] W. O'Reilly, Rock and Mineral Magnetism, Blackie and Son, New York, 1984, 220 pp.
[14] J. Riisager, M. Perrin, P. Rochette, Paleointensity results from Ethiopian basalts: implications for the Oligocene geomagnetic field strength, J. Geophys. Int. 138 (1999) 590–596.
[15] H.P. Johnson, W. Lowrie, D.V. Kent, Stability of anhysteretic remanent magnetization in fine and coarse magnetite and maghemite particles, Geophys. J. R. Astron. Soc. 41 (1975) 1–10.
[16] J.L. Kirschvink, The least squares line and plane and the analysis of paleomagnetic data, Geophys. J. R. Astron. Soc. 62 (1980) 699–718.
[17] P.L. McFadden, M.W. McElhinny, Classification of the reversal test in paleomagnetism, Geophys. J. Int. 130 (1990) 725–729.
[18] A. Schult, Paleomagnetism of Tertiary volcanic rocks from the Ethiopian Southern Plateau and the Danakil block, J. Geophys. 40 (1974) 203–212.
[19] P.L. McFadden, R.T. Merrill, M. McElhinny, S. Lee, Reversals of the Earth's magnetic field and temporal variations of the dynamo families, J. Geophys. Res. 96 (1991) 3923–3933.
[20] T. Kidane, V. Courtillot, I. Manighetti, L. Audin, P. Lahitte, X. Quidelleur, P.Y. Gillot, Y. Gallet, J. Carlut, T. Haile, New paleomagnetic and geochronologic results from Ethiopian Afar: block rotations linked to rift overlap and propagation, and determination of a $\sim$2Ma reference pole for stable Africa, J. Geophys. Res. (2002) in press.
[21] J. Carlut, V. Courtillot, How complex is the time-averaged field over the past 5 million years?, Geophys. J. Int. 134 (1998) 527–544.